

CYVANTA

Technical Documentation — Cybersecurity Services Website

Project Title: CYVANTA — Cybersecurity Services

Document purpose: technical documentation for the completed frontend task, following the heading structure and submission style requested in the supplied Travel and Hospitality documentation reference.

1. Introduction & System overview:

CYVANTA is a premium multi-page cybersecurity services website designed to present security capabilities, services, insights and contact workflows through a modern dark interface. The frontend combines semantic page structure, responsive layouts, client-side interactions, authentication demonstrations, role-based dashboard routing and motion effects.

The application is implemented as a Multi-Page Application (MPA). Public pages and dashboard views are independent HTML documents, while shared CSS and JavaScript provide consistent navigation, validation, animation and responsive behavior.

2. Technology Stack & Dependencies

Core Technologies:

- HTML5: semantic structure for public pages, authentication screens, dashboards and the custom error page.
- CSS3: responsive layout, design tokens, glassmorphism surfaces, gradients, hover states, navigation states and dashboard styling.
- Vanilla JavaScript (ES6+): navigation behavior, mobile menus, authentication validation, role-based routing, dashboard identity rendering and CTA handling.
- GSAP: page entrance motion, scroll-linked motion, card reveals, interactive hover motion and high-performance transforms.
- AOS (Animate On Scroll): staggered viewport entrance animations across content sections and dashboard visuals.

External Libraries & CDNs:

- GSAP 3.12.5 CDN
- GSAP ScrollTrigger, Observer and Flip modules where loaded by dashboard pages
- AOS 2.3.4 CDN
- Google Fonts resources used by the visual system

3. File Architecture & Directory Structure

```
CYVANTA/  
■■■ index.html  
■■■ about.html  
■■■ services.html  
■■■ blog.html  
■■■ contact.html  
■■■ login.html  
■■■ signup.html  
■■■ 404.html
```

```
■■■ admin/
■ ■■■ dashboard-overview.html
■ ■■■ dashboard-security.html
■ ■■■ dashboard-activity.html
■ ■■■ dashboard-reports.html
■ ■■■ dashboard-settings.html
■■■ client/
■ ■■■ dashboard-overview.html
■ ■■■ dashboard-security.html
■ ■■■ dashboard-activity.html
■ ■■■ dashboard-reports.html
■ ■■■ dashboard-settings.html
■■■ CSS/ style.css
■■■ JS/ main.js
■■■ assets/ images + icons
■■■ documentation/ CYVANTA_Technical_Documentation.pdf
```

The structure separates public pages, role-specific dashboards, shared styling, behavior, visual assets and technical documentation. Redundant dashboard redirect wrapper files from earlier iterations are not part of the final source because login.js routes directly to the role dashboard entry pages.

4. Visual Design System (UI/UX)

- Glassmorphism-inspired surfaces: translucent dark cards, subtle borders and controlled blur create depth without reducing readability.
- Flexbox-first responsive composition: navigation, cards, dashboard panels and content groups adapt across mobile, tablet and desktop widths.
- Interactive micro-animations: hover elevation, scale, pointer movement, reveal transitions and active-state feedback are used on interactive components.
- Visual security language: cyan/blue light, dark surfaces, shield/network imagery, technical labels and compact uppercase metadata reinforce the cybersecurity identity.
- Fixed dashboard sidebar: role-specific navigation remains stable on desktop and converts to a controlled hamburger drawer on smaller screens.

5. Visual Design Language & Color Palette

Color / Token	Value	Usage
Background	#040711	Primary page background
Background 2	#080d18	Secondary dark surfaces
Surface	#0b1220	Cards and panels
Text	#f4f8fc	Primary headings and text
Muted	#8e9bad	Secondary descriptions
Cyan	#55e6ff	Primary interactive accent / CTA
Cyan 2	#18b9d6	Secondary cyan states
Violet	#8c7bff	Secondary visual accent
Lime	#b8ff4a	Selective highlight
Green	#58e6a8	Positive/security state
Danger	#ff6b8a	Error / threat state

Authentication buttons use a cyan filled Sign Up treatment and dark outlined Login treatment. Semantic feedback uses green for successful states and red/pink for invalid or threat-related states.

Overall Layout:

Public-facing pages use a consistent header/footer system, page hero or section introduction, content modules, imagery and call-to-action areas. Dashboard pages use a fixed sidebar, top header with portal identity and user email/avatar, followed by page-specific content.

6. Public-Facing & Internal Architecture Overview

A. Public-Facing Architecture

Home Page (index.html): Hero security statement, capability cards, lifecycle/process content, proof/metrics, security imagery and final CTA.

About Page (about.html): CYVANTA positioning, security philosophy, operating approach and supporting visual modules.

Services Page (services.html): Cybersecurity capability/service groups with interactive cards and visual content.

Insights Page (blog.html): Security insight/article cards with imagery, metadata and hover interactions.

Contact Page (contact.html): Inquiry form, contact/support details and client-side validation.

Authentication (login.html & signup.html): Role-aware login, account creation validation, password visibility control and visual security imagery.

Error Page (404.html): Branded fallback page with clear Back to Home navigation.

B. Internal Architecture (Dashboards):

Admin Dashboard

- Overview — posture score, threat activity and control health.
- Security Operations — security controls and threat-response context.
- Users & Assets — activity and protected-environment information.
- Reports — posture, compliance and priority-risk summaries.
- Settings — account and portal preferences.

Client Dashboard

- Overview — client security posture at a glance.
- Protection — protection controls and monitored security context.
- Activity — recent security events and response timeline.
- Reports — client-facing posture and risk summaries.
- Settings — client account preferences.

Dashboard tabs are separate HTML views in the final MPA implementation. Each page highlights its active navigation item and keeps content specific to that view rather than placing every dashboard module on the Overview page.

7. Authentication & Error Handling

Client Side Security & Validation:

- Email validation uses native email validity checks.
- Login requires a selected Admin or Client role.
- Login requires a minimum 8-character password.
- Signup validates name format, email, password complexity, password confirmation and policy acceptance.
- Password visibility is controlled through an eye-style toggle button.
- Invalid fields display targeted inline error guidance.

Role-Based Routing:

After valid login, the selected role determines the dashboard destination: Admin routes to `admin/dashboard-overview.html` and Client routes to `client/dashboard-overview.html`. Dashboard pages verify that the role matches the portal type before rendering protected content; invalid or missing identity information redirects back to login.

Session identity:

The frontend demonstration stores the normalized email and role in `sessionStorage/localStorage` so dashboard pages can display the logged-in email, avatar initial and role label. This is a frontend demonstration only and is intended to be replaced by server-side authentication in production.

Error handling:

Unknown or unavailable navigation destinations are routed to the custom 404 page where applicable. Authentication validation prevents invalid form submission and gives immediate user feedback.

8. Responsive & Performance Methodologies

- Mobile: below 768px with compact navigation and hamburger menu.
- Tablet: 768px–991px with hamburger navigation, centered menu items and side-by-side authentication actions.
- Desktop: 992px and above with full navigation and desktop header actions.
- Fluid widths, max-width containers, flex layouts and overflow protection prevent horizontal page stretching.
- Fixed mobile/tablet menu behavior prevents background scrolling while the navigation drawer is open.
- GSAP transforms and opacity-based motion are preferred for smoother animation performance.
- AOS provides lightweight scroll entrance effects and GSAP handles more controlled motion sequences.
- Images use modern WebP assets and non-critical dashboard imagery uses lazy loading.

9. SEO & Asset Optimization

- Semantic HTML elements are used across headers, navigation, main content, sections, forms, sidebars and footers.
- Pages include title and description metadata and responsive viewport metadata.
- robots.txt and sitemap.xml are included for deployment preparation.
- Cybersecurity visual assets are stored under assets/images/cyber/ to keep media organized by domain.
- Optimized WebP imagery is used for the photographic cybersecurity assets; the final WebP assets are kept below the requested 100 KB target.
- Images that are not immediately required are marked loading="lazy" where appropriate.
- The Stackly/CYVANTA logo is kept as a transparent WebP asset for consistent use on dark UI surfaces.

10. Future Scalability Insights

Backend Framework Integration:

The static HTML/JavaScript implementation can be migrated to React, Angular or another component framework to centralize reusable UI components, routing and application state.

Database & Vector Storage:

Frontend-only identity and demonstration metrics can be replaced by a secure relational database and, where appropriate, vector storage for security knowledge retrieval and intelligent search use cases.

WebSocket Integration:

A production security portal can replace static activity displays with WebSocket or server-sent real-time streams for threat events, alert status, analyst actions and live security telemetry.

Final Implementation Note

This document describes the final CYVANTA frontend source structure and behavior. It intentionally follows the heading organization of the supplied reference document while using CYVANTA-specific terminology and implementation details.